

WATER RESOURCES

Water is the main constituent of hydrosphere and is a renewable resources. About $\frac{3}{4}$ of the earth surface is occupied by oceans which contain about 97.5% of the earth's water in strongly saline condition and the rest 2.5% is fresh water.

Division of Fresh water	
1.97%	Polar or glacial ice
0.5%	Groundwater
0.02%	Lakes & Rivers
0.01%	Soil
0.0001%	Atmosphere

Water cycle play an important role in maintaining different forms of water in nature. Water passes into the atmosphere by evaporation from the surface of moist earth, lakes, streams, oceans etc. Water vapour gets condensed and returns to the earth in the form of rain, hail and snow. There is a circulation of water from sea to land and back.

Forms of Fresh water

Fresh water mainly occurs in two forms: Groundwater and Surface water

1. Groundwater: The groundwater is contained in aquifers. An aquifer is a highly permeable layer of sediment or rock containing water.

2. Surface water: Surface water occurs in the forms of streams, rivers, lakes, ponds, etc. It comes through precipitation (rain fall, snow).

Effects of Overuse of Fresh water:

- **Lowering of water table** – Excessive use of groundwater for drinking, irrigation and domestic purposes has resulted in rapid depletion of groundwater table and drying of wells.
- **Ground subsidence** – When groundwater withdrawal is more than its recharge rate, the sediments in the aquifer become compacted. It results in sinking of land which may damage buildings, reverse the flow of sewers and tidal flooding.

Problems related with water resources

1. About 40% of the world's population lives in arid or semi-arid region. These people have to spend substantial amount of time, energy and effort in obtaining water for their domestic and agricultural use.
2. To meet the requirements of huge population, surface waters are over drawn, resulting in drying up of nearby wet lands.
3. The groundwater may also dry out, when more groundwater is removed for human use than can be recharged by rainfall or snowmelt.
4. Excessive irrigation in semi-arid and arid regions can cause salt accumulation in the soil, which may reduce crop productivity.
5. The continuous depletion of groundwater along the coastal regions often leads to the movement of saline sea water into fresh water wells spoiling their water quality.
6. Heavy rainfall on exposed soils results in rapid runoff causing soil erosion.

Conservation and Management of water

1. Increase in irrigation efficiency in agricultural fields through sprinkle irrigation, drip irrigation etc. and by reducing water wastage.
2. Recycling of used water in industries so as to reduce water wastage.
3. Reduction in domestic water wastage by constructing waste water treatment plants and recycling the treated water.
4. Harvesting of rainwater by adopting practices like storing of rainwater and recharging groundwater.
5. Protection of water sheds and afforestation to improve water economy.
6. Construction of dams and water reservoirs to control floods, ensure year round supply of water and generation of electricity.
7. Desalinization of sea water and saline groundwater to make it fit for human consumption and other purposes.
8. Diversion of water bodies through canals to increase water supply in drier areas.
9. Regular dredging and desiltation of rivers, streams and other water bodies.